

Supplementary Materials for “Language Embeddings Meet Shallow Autoencoders”

A BASELINES

As baselines to compare our models, we considered the following:

- **Random Baseline:** This baseline recommends items for each user by selecting them uniformly at random.
- **Inductive Matrix Factorization (IMF)** [15, 48, 66]: Inductive matrix factorization is a family of learning models widely used for leveraging features while factorizing a sparse observed matrix. Based on [66], we factorize matrix $X = UMS^T$ with predictor $\hat{X} \approx UMS^T$. The rank constraint imposed by the hyperparameter d ($M \in \mathbb{R}^{d \times r}$) was cross-validated in the range $\{10, 20, 50, 100\}$.
- **Nearest Neighbor-based Methods (NN) and Content-based Recommendation (CBR):** Algorithms based on nearest neighbors are widely used for content-based filtering, and because of that, also for recommending cold-start items [3, 32, 45]. NN and CBR first train a collaborative filtering method only considering warm-start items, returning the estimated matrix $\hat{X}$. In the prediction step, NN then uses the side information to find the most similar warm-item to the cold item for which the recommendation is requested. The output score for the cold-start item is copied to the most similar warm item. On the other hand, CBR computes a warm-start to cold-start matrix similarity $M \in \mathbb{R}^{n \times \tilde{n}}$, with entry $M_{i,j}$ being the side information cosine similarity between warm-start item- i and cold-start item j . The predictor is $\hat{X} = \hat{X}M$. We use a collaborative filtering method [33] as the backbone, implemented with the Pytorch backend.
- **Heater** [44]: Heater is a model engineered for the cold-start problem, combining separate training and joint-training frameworks to extract feature representations for cold-start users and items. We implement the model with Pytorch as backend and only considering side information for items.
- **MTPR** [6]: MTPR is a model that optimizes dual item representations, incorporating both normal and counterfactual representations, aiming to address the training-testing gap for cold-start items. Hyperparameter selection is done according to Section 4.1.4 of the original paper².
- **CCFCRec** [42]: A contrastive learning-based framework proposed for addressing cold-start recommendation challenges. This framework leverages co-occurrence collaborative signals within warm training data to mitigate the problem of unclear collaborative embeddings for recommending cold-start items. The experimental setting follows Section 4.1 of the original paper and the implementation adapts the source code of the authors³.
- **TEASER** [5]: A hybrid shallow autoencoder-based recommender system that combines the strengths of collaborative and content-based filtering. It utilizes implicit feedback interaction data to learn item similarities from attributes. The method is particularly suited for addressing the cold-start problem, as it can leverage item metadata alone to make recommendations in the absence of sufficient user interaction

²Original code: <https://github.com/duxy-me/MTPR>

³Original code: <https://github.com/zzhin/CCFCRec>

Table A-1: Summary of the used datasets’ statistics after pre-processing.

Dataset	# users	# items	interactions	density
MovieLens	136 677	20 720	9.99 M	0.35 %
GoodBooks	53 366	10 000	4.12 M	0.77 %
Amazon Electronics	73 411	33 126	1.40 M	0.05 %

data. We implement the model with PyTorch as the backend, considering only side information for items.

- **LAE (Linear Autoencoder)** [19]: A simple linear autoencoder for ablation without any constraints or diagonal gating.

We note that recent prompting-based methods and transformer fine-tuning approaches (e.g., [24, 36, 37]) do not yet scale to the large user and large catalog settings considered here, and are therefore not directly comparable. For our model, the hyperparameter $\lambda \in \{10^{-2}, 10^{-1}, \dots, 10^5\}$ was selected.

B DATASETS

We will briefly describe the datasets used for evaluation, which are widely recognized as benchmark datasets for recommender system evaluation: Amazon Electronics, Goodbooks, and MovieLens. All the datasets were originally collected with explicit feedback, e.g., one-to-five star user-item ratings. Following a standard procedure in recommender systems research [55], we consider only ratings of four or higher as positive samples of interactions, dropping the others. A short summary of dataset is available in Table A-1.

- **Amazon Electronics:** This dataset belongs to the online shopping domain [20]. We preprocessed the dataset to include only users and items with at least ten interactions. Since the original dataset already contains item titles and descriptions, we encoded them into vectors using the sentence transformers library [62] and used them as side information.
- **GoodBooks** [40]: Goodbooks belongs to the book recommendation domain. To get side information for the items, we used available book descriptions (we refer to [47, 50, 61]) and encoded them, along with the title, using models from the sentence transformers library [62].
- **MovieLens (ML-20M)** [9]: For the MovieLens dataset, we only keep users with at least five interactions. For side information, we collected additional data for movies in the dataset from [57] and [58]: we extracted categorical features such as actors, director, year of release, and language, which we combined with genres from the original dataset.

Nowadays, the use of embeddings based on large-language models is pervasive, with a large number of model options available. To select a model for generating embeddings for our analysis, we first

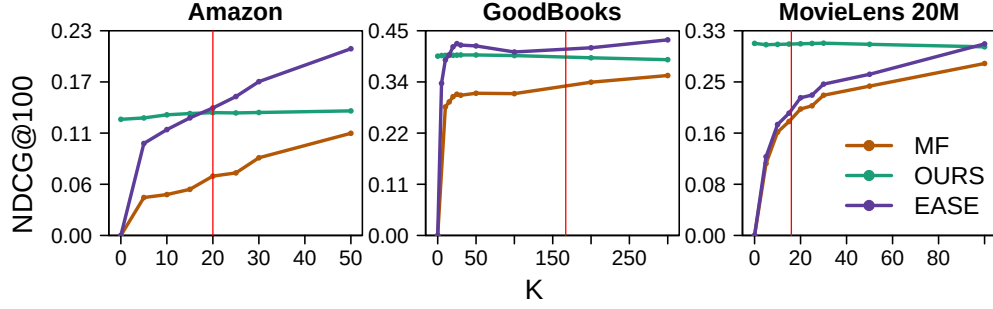


Figure A-1: NDCG@100 performance on the transition from cold-to-warm status recommendation organized by datasets.

selected three popular models: *all-MiniLM-L12-v2*⁴, *all-mpnet-base-v2*⁵, and *distiluse-base-multilingual-cased-v2*⁶ from the sentence transformers [62] library and evaluated them with CBR. We observed that *all-mpnet-base-v2* consistently outperformed its counterparts. Therefore, we used this model to create the embeddings that serve as side information for all other models.

Train-Validation-Test Split: For each dataset, we randomly choose 1,000 items for validation and 1,000 items for testing, and remove all interactions associated with these items from the training set. The remaining interactions are used for training.

C COLD-TO-WARM START TRANSITION:

Our main experimental focus is on cold-start items with no prior interactions observed. However, it is important to analyze how our method behaves during the transition phase, i.e., when a few new interactions for the cold-start items are received. To do this, we proceed as follows: for each cold-start item in the test set, we randomly select K interactions and add them to the training set, and finally train the models once more. The remaining interactions are still in the test set. By varying K , we compute NDCG@100 for our model, treating the items with additional observations as warm items. In this scenario, pure collaborative filtering methods are also applicable. Accordingly, we add the shallow autoencoder EASE [29], as well as traditional matrix factorization [11] for comparison purposes.

The results are presented in Figure A-1. The vertical red line in the graphs outlines the median number of interactions an item had in the original training set. We observe that our model maintains stable recommendation metrics when changing K , while pure collaborative methods significantly improve their performance as the number of observed interactions increases. For instance, EASE outperforms our model across all datasets after observing enough interactions. Interestingly, for the Amazon and Goodbooks datasets, our model outperforms matrix factorization throughout the warming scenario.

D IMPLEMENTATION DETAILS AND COMPLEXITY ANALYSIS

Our ALS procedure optimizes $\hat{\mathbf{e}}_i$ by alternating over i in

$$\hat{\mathbf{e}}_i = \left(n_i \begin{bmatrix} S^{-i} \end{bmatrix}^\top \begin{bmatrix} S^{-i} \end{bmatrix} + \lambda I_d \right)^{-1} \begin{bmatrix} S^{-i} \end{bmatrix}^\top (\mathcal{X}^{-i})^\top \mathbb{1}_{n_i}, \quad (\text{A-1})$$

while fixing $\mathcal{E}^{-i}$. Although, in principle, co-occurring items should not be updated simultaneously (to avoid information leakage through shared context), we observe empirically that precomputing some quantities from the initial values and updating small batches of items in parallel can significantly speedup convergence without degrading performance. Our model was trained solely on CPUs using an implementation in NumPy, while deep-learning baseline models were trained on a DGX A100 GPU card.

There are several ways to verify the convergence of our algorithm. For instance, the algorithm can rely on any of the following stopping criteria: (1) repeating the updates for a fixed number of iterations; (2) stopping when the training or validation loss changes by less than a prescribed threshold; or (3) storing the previous value of $\mathcal{E}$ and stopping when $\mathcal{E}$ does not change significantly. Following [56], for the last option we use a threshold $0 < \tau \in \mathbb{R}$ and stop when

$$\frac{\|\mathcal{E} - \mathcal{E}_{\text{old}}\|_{\text{Fro}}}{\|\mathcal{E}\|_{\text{Fro}}} < \tau.$$

Computational Complexity: Two main operations are needed when computing $\mathbf{e}_i$: (i) computing the dense Gram matrix multiplication $\begin{bmatrix} S^{-i} \end{bmatrix}^\top \begin{bmatrix} S^{-i} \end{bmatrix}$ and (ii) inverting the augmented Gram matrix $(n_i \begin{bmatrix} S^{-i} \end{bmatrix}^\top \begin{bmatrix} S^{-i} \end{bmatrix} + \lambda I_d)$. The first operation has complexity $O(d^2 n)$ and the second has complexity $O(d^3)$. Since generally $d \ll n$, in practical applications we can assume that the full complexity of the algorithm is $O(Nd^2 n^2)$ if this operation is performed N times for each of the n warm-start items.

Note, however, in terms of memory, we only need to store the essentials for computation, with space complexity $O(M + nd)$, where M is the number of observed entries in X , and nd accounts for storing $\mathcal{E}$ and S . We can significantly reduce the time complexity by increasing memory usage via precomputation. Note that, when computing (A-1), the only term that changes across updates is $\mathcal{E}^{-i}$. By regrouping terms and precomputing the quantities that do not depend on the current iterate, we reduce the per-update cost and obtain an overall time complexity of $O(Nn)$, at the expense of increasing the memory complexity to $O(nd^2)$.

⁴<https://huggingface.co/sentence-transformers/all-MiniLM-L12-v2>

⁵Available in: <https://huggingface.co/sentence-transformers/all-mpnet-base-v2>

⁶Available in: <https://huggingface.co/sentence-transformers/distiluse-base-multilingual-cased-v2>

E LLM-GUIDED TOPIC DICTIONARY CONSTRUCTION / CLUSTER LABELING

We use an LLM to construct a topic dictionary for interpreting user representations and cluster centroids with human-readable semantic descriptors. Specifically, we used OpenAI's GPT-5.2 Pro (gpt-5.2-pro) to generate the topic sets and vocabularies described above, using as input the book synopses (one per line) and a fixed-format prompting template.[22] The prompt below is provided together with the book synopses and instructs the LLM to output a set of topics, each paired with a synopsis-grounded vocabulary. We apply the same prompting procedure to MovieLens by replacing the book synopses with movie plot summaries and using an otherwise identical prompt template.

Prompt (verbatim).

You are helping me build a topic dictionary for semantic interpreting representations in a book recommendation system.

Input: I will provide a text file where each line contains ONE book in the format:

TITLE: SYNOPSIS

(English text; some lines may contain non-English snippets.)

Task: Create a hierarchical topic dictionary.

Each topic must:

- 1) be specific (not generic like "drama", "interesting", "story", "life"),
- 2) include AT LEAST 20 representative vocabulary items (single words or short phrases, max 3 words),
- 3) focus on semantic content that would plausibly trigger user preference (themes, settings, plot motifs, character archetypes),
- 4) avoid author names, book titles, character names, and franchise/series references.

Topic design rules:

- Topics should be mutually distinguishable: minimize overlap in vocab between topics.
- Do NOT use near-duplicate topics

Vocabulary rules:

- Provide at least 20 items per topic.
- Use lowercase for all vocabulary items.
- Avoid stopwords and ultra-common words ("the", "and", "good", "thing", "people").
- Prefer terms likely to appear in synopses over abstract adjectives ("beautiful", "deep").

Output format:

```
"topic_name_1": ["w1", "w2", ...],
"topic_name_2": ["w1", "w2", ...],
...
"topic_name_N": ["w1", "w2", ...]
```

For cluster labeling, we used the following prompt:

Prompt (verbatim).

You are given 15 clusters that represents user taste in a book recommendation system. For each cluster, I provide:

- cluster id (0..14)
- a list of five topics entries (already ranked; first is most important)

Your task:

Create ONE concise cluster label per cluster.

Rules:

- 1) The label must reflect the COMBINATION of themes, not just the top one.
- 2) Do NOT use franchise/series names, character names, author names, or specific IPs.
- 3) Prefer preference-triggering semantics: settings, motifs, subgenres, archetypes.
- 4) Labels must be short and punchy (2-7 words).
- 5) Avoid generic labels like "Mixed", "Various", "Drama", "Action", "Comedy" alone.
- 6) When a cluster mixes two worlds (e.g., "romance + music"), explicitly fuse them.

Output format:

```
"0": "<label>",
"1": "<label>",
...
"14": "<label>"
```

Bellow we enumerate the Topic-dictionary based on GoodBooks:

- (1) **fantasy_magic_general**: fantasy, magic, magical, wizard, wizards, witch, witches, sorcery, sorcerer, spell, spells, wand, wands, potion, potions, enchantment, enchanted, curse, cursed, prophecy, prophecies, myth, myths, legend, legends, dragon, dragons, elf, elves, dwarf, dwarves, orc, orcs, goblin, goblins, troll, trolls, giant, giants, faerie, faeries, fairy, fairies, kingdom, throne, castle, tower, realm, quest, journey, adventure, hero, heroine, villain, dark, darkness, ancient, artifact, amulet, ring, sword, blade, dagger, shield, battle, warrior, knight, mage, charm, rune, runic, portal, worlds, otherworld, destiny, fate, ritual, summon, summoning, spellcasting, incantation
- (2) **magic_school_wizardry**: hogwarts, school, boarding, dorm, castle, great, hall, professor, headmaster, student, students, classroom, lessons, homework, library, common, room, house, gryffindor, slytherin, ravenclaw, hufflepuff, quidditch, broom, brooms, broomstick, wand, wands, spell, spells, potion, potions, cauldron, cloak, owl, owls, letter, letters, platform, train, express, sorting, hat, muggle, muggles, wizarding, witchcraft, wizardry, defense, dark, arts, horcrux, azkaban, dementor, dementors, patronus, phoenix, goblet, triwizard, tournament, chamber, secrets, stone, sorcerer, sorcerer's, deathly, hallows, voldemort, dumbledore, hermine, ron, harry
- (3) **epic_quest_middleearth**: middle-earth, middleearth, shire, hobbit, hobbits, bilbo, frodo, gandalf, ring, rings, sauron, mordor, cracks, doom, mount, journey, quest, fellowship, tower, towers, return, king, kings, dark, lord, shadow, orc, orcs, uruk, elf, elves, dwarf, dwarves, ranger, wizard, sword, bow, arrow, battle, siege, fortress, mountain, mines, cave,

- caves, dragon, smaug, treasure, kingdom, throne, crown, artifact, power, bind, rule
- (4) **mythology_gods_heroes:** mythology, myth, greek, roman, gods, god, goddess, olympian, olympians, zeus, poseidon, hades, athena, ares, apollo, artemis, hera, demeter, persephone, titan, titans, kronos, demigod, demigods, half-blood, camp, oracle, prophecy, quest, labyrinth, underworld, monster, monsters, hydra, minotaur, cyclops, satyr, nymph, hero, heroes, odyssey, odysseus, ithaca, troy, trojan, muse, epic, voyage, sea, fleece, trident, lightning, thief
- (5) **dragons_riders:** dragon, dragons, hatchling, hatch, egg, rider, riders, dragonrider, riding, saddle, fire, flame, scales, wing, wings, claw, claws, breath, flight, sky, sword, ancient, empire, king, tyrant, evil, dark, enemy, prophecy, destiny, magic, spell, elders, legend, training, mentor, quest, battle, war
- (6) **supernatural_creatures:** vampire, vampires, werewolf, werewolves, coven, blood, thirst, bite, fang, fanged, undead, immortal, immortality, sunlight, moon, full, curse, cursed, shapeshift, shapeshifter, pack, hunter, hunters, demon, demons, angel, angels, spirit, spirits, ghost, ghosts, haunted, possession, possessed, witch, warlock, fae, shadowhunter, shadowhunters, underworld, darkness, monster, monsters, supernatural, occult
- (7) **dystopia_totalitarian:** dystopia, dystopian, utopia, utopian, totalitarian, authoritarian, regime, state, party, propaganda, surveillance, spy, spies, telescreen, thoughtcrime, ministry, truth, police, secret, control, oppression, oppressed, rebellion, rebel, rebels, resistance, uprising, revolution, district, capitol, faction, virtue, caste, hierarchy, conditioning, conformity, censorship, ban, banned, burn, burning, books, book-burning, handmaid, gilead, commander, womb, fertility, ovaries, matching, society, rules, punishment, fear, indoctrination
- (8) **post_apocalyptic_survival:** apocalypse, apocalyptic, postapocalyptic, post-apocalyptic, end, world, plague, pandemic, virus, outbreak, infection, collapse, ruins, ash, burned, wasteland, desolate, barren, survive, survival, scavenge, scavenging, supplies, ration, food, water, shelter, bunker, refuge, camp, gun, pistol, raid, raiders, bandits, lawless, horde, death, starvation, cold, winter, storm, evacuation, disaster, catastrophe, catastrophic, rescue, signal, alone, isolation, dark, road, journey
- (9) **science_fiction_tech:** science, scientific, technology, technological, engineer, engineering, mechanical, machine, machines, robot, robots, satellite, space, spaceship, astronaut, mission, mars, planet, galaxy, interstellar, future, futuristic, laboratory, lab, experiment, experiments, research, facility, genetic, genetics, dna, clone, cloning, mutation, modified, bioengineering, reproductive, ecology, calorie, agriculture, calories, data, network, computer, code, cipher, cryptography, algorithm, simulation, virtual, time, travel, tesseract
- (10) **time_travel_timeline:** time, travel, time-travel, timeline, past, future, present, paradox, chronological, chronology, displacement, chrono, tesseract, dimension, fifth, space-time, loop, reset, memory, flashback, history, era, century, years, decades, calendar, date, sequence, rewind, forward
- (11) **mystery_detective_investigation:** mystery, mysterious, detective, investigation, investigate, investigating, clue, clues, evidence, suspect, suspects, alibi, case, cold, missing, disappearance, kidnapped, abduction, murder, killed, killer, crime, criminal, forensic, police, inspector, journalist, reporter, search, hunt, track, trail, riddle, puzzle, secret, secrets, hidden, reveal, truth, lie, lies, deceit, conspiracy, plot, twist, thriller, suspense
- (12) **conspiracy_codes_symbols:** conspiracy, secret, society, societies, illuminati, priory, sion, vatican, church, catholic, louvre, curator, murdered, symbol, symbols, symbologist, cryptic, cipher, code, codes, decode, decipher, riddle, riddles, puzzle, puzzles, manuscript, ancient, history, artifact, vault, crypt, crypts, catacomb, catacombs, cathedral, cathedrals, mystery, clues, trail, hidden, painting, da, vinci, leonardo
- (13) **crime_legal_courtroom:** law, legal, lawyer, attorney, defense, prosecutor, judge, jury, trial, court, courtroom, case, evidence, testimony, witness, sentence, conviction, prison, inmate, penitentiary, death, row, execution, electric, chair, appeal, justice, injustice, racism, racial, violence, crime, murder, assault, rifle, sniper, mob, mafia, fbi, investigation
- (14) **romance_relationships:** romance, romantic, love, lovers, kiss, kissing, passion, desire, lust, affair, flirtation, courtship, dating, boyfriend, girlfriend, fiancé, fiancée, wedding, marriage, husband, wife, relationship, breakup, heartbreak, jealousy, betrayal, temptation, seductive, intimacy, soulmate, devotion, tender, sweet, forever, letters, notebook
- (15) **erotic_billionaire_romance:** erotic, sex, sexual, sensual, kink, bondage, dominant, submissive, submission, control, obsession, desire, pleasure, ecstasy, intimate, intimacy, contract, rules, safe, word, billionaire, wealth, rich, luxury, penthouse, limousine, helicopter, multinational, business, entrepreneur, tortured, demons, possess, possession
- (16) **family_friendship_belonging:** family, parents, mother, father, daughter, son, sister, brother, home, house, marriage, divorce, widow, widowed, orphan, orphaned, adopt, adoption, foster, child, children, friend, friends, friendship, best, sisterhood, bond, bonding, loyal, loyalty, community, town, neighbor, neighbors, support, group, care, protect, protection, love, forgiveness, reconciliation
- (17) **coming_of_age_school:** coming-of-age, coming_of_age, teen, teens, teenage, teenager, teenagers, adolescent, adolescence, youth, young, adult, young-adult, ya, highschool, high-school, middle-school, boarding-school, college, campus, dorm, dormitory, class, classes, teacher, teachers, principal, headmaster, bully, bullies, friend, friends, friendship, clique, outsider, misfit, loner, crush, first-love, romance, dating, prom, party, graduation, identity, self-discovery, self-esteem, insecurity, confidence, peer, pressure, coming, out, rebellion, rules, detention, sports, team, club
- (18) **classics_society_manners:** regency, victorian, manners, society, gentleman, lady, estate, manor, ball, dance, courtship, marriage, proposal, pride, prejudice, wit, satire, austere, fortune, inheritance, money, status, class, upper, middle, poor, rich, scandal, gossip, reputation, virtue, honor, duel, tragedy
- (19) **war_holocaust_occupation:** war, world, battle, soldier, army, air, forces, bomber, prisoner, camp, concentration, auschwitz, buchenwald, nazi, germany, hitler, occupation, occupied, resistance, underground, escape, hiding, attic, annexe, gestapo,

- jewish, holocaust, refugee, bomb, bombing, firebombing, dresden, front, line, uniform, veteran, patriot, flag
- (20) **historical_revolution_empire**: revolution, french, paris, london, guillotine, mob, uprising, monarchy, king, queen, empire, imperial, feudal, castle, cathedral, monk, monastery, plague, poverty, injustice, prison, convict, escape, revenge, treasure, isle, count, marquis, baron, duke, court, soldiers, march, battlefield
- (21) **social_justice_race_poverty**: racism, racial, segregation, civil, rights, injustice, justice, law, trial, court, jury, southern, alabama, mississippi, poverty, poor, working, class, wages, minimum, wage, wal-mart, waitress, maid, hotel, service, industry, labor, sharecropper, okies, dust, bowl, migrant, immigrant, refugee, slavery, runaway, slave, lynch, klan, violence, activism, protest
- (22) **memoir_biography_true_story**: memoir, biography, autobiography, diary, journal, true, story, real, life, childhood, family, growing, up, author, writer, recount, account, testimony, remember, memories, reminiscence, survival, resilience, redemption, trauma, poverty, homeless, addiction, alcohol, abuse, escape, journey, travel, adventure, mountain, wilderness, hitchhiked, alaska, olympics, entrepreneur, inventor, icon, history
- (23) **philosophy_ethics_existential**: philosophy, philosophical, ethics, ethical, moral, morality, virtue, justice, duty, conscience, guilt, sin, redemption, forgiveness, truth, meaning, absurd, existential, existence, life, death, mortality, suffering, freedom, choice, responsibility, reason, rational, logic, utility, utilitarianism, apology, socrates, plato, argument, debate, speech, trial, human, nature, soul
- (24) **religion_spirituality_faith**: god, faith, prayer, pray, church, bible, scripture, christian, catholic, baptist, minister, preacher, sermon, angel, angels, heaven, hell, salvation, redeem, redemption, sin, forgiveness, miracle, spiritual, spirituality, soul, souls, divine, holy, saint, devotion, worship, prophecy, apocalypse, rapture, afterlife
- (25) **horror_thriller_fear**: horror, terror, terrifying, nightmare, spooky, creepy, sinister, evil, haunted, ghost, ghosts, demon, demons, possession, cursed, curse, blood, murder, killer, death, dead, monster, monsters, beast, beasts, dark, shadow, violence, panic, fear, scream, madhouse, asylum, hotel, overlook, winter, storm, isolation, cabin, woods, attic, shivers
- (26) **mental_health_hospital**: mental, health, depression, anxiety, madness, insane, insanity, therapy, therapist, psychiatric, hospital, ward, nurse, doctor, diagnosis, patient, patients, medication, shock, electroshock, trauma, ptsd, suicide, self-harm, grief, mourning, loss, addiction, alcohol, drink, drinking
- (27) **illness_cancer_medical**: cancer, tumor, leukemia, illness, disease, diagnosis, terminal, treatment, chemo, chemotherapy, radiation, surgery, transfusion, shots, hospital, doctor, nurse, clinic, medicine, medical, miracle, support, group, patient, pain, ovarian, blood, bone, marrow, donor, genetic, syndrome, down
- (28) **travel_adventure_journey**: journey, travel, trip, voyage, road, walk, hitchhike, train, bus, plane, ship, boat, island, desert, mountain, river, sea, ocean, wilderness, forest, north, south, paris, rome, london, scotland, afghanistan, kabul, bangkok, alaska, america, york, city, escape, adventure, expedition, quest, odyssey, pilgrimage
- (29) **books_reading_writing_storytelling**: book, books, reading, read, reader, story, stories, tale, tales, novel, chapter, chapters, author, writer, writing, literature, classic, poetry, poem, poems, diary, journal, letters, notebook, library, librarian, school, teacher, study, studying, text, translation, edition, illustrations, illustrated, words, language, prose, narrator, narration, essay, essays
- (30) **children_picture_books_rhymes**: children, child, kids, kid, baby, mother, father, family, bedtime, sleep, nursery, picture, book, illustrations, rhymes, rhyme, funny, silly, wacky, cat, hat, mouse, goat, tree, eggs, ham, bear, lion, spider, web, pig, barn, zoo, puppy, duck, crocodile, whale, unicorn, shadow, garbage, play, playful, school, principal, teacher, story, tale
- (31) **animals_wilderness_nature**: animal, animals, wild, wilderness, forest, woods, mountain, river, sea, ocean, island, desert, arctic, snow, winter, storm, tiger, bear, lion, wolf, dog, puppy, horse, elephant, spider, rat, rats, bee, bees, honey, pig, farm, barn, web, zoo, circus, train, hunt, hunting, survival, pack, herd, nature, outdoors
- (32) **music_art_performance**: music, musical, song, songs, guitar, strings, band, concert, sing, singing, record, records, album, piano, composer, artist, art, painting, portrait, museum, gallery, theatre, theater, play, drama, performance, stage, actor, actress, circus, acrobats, magicians, magic, show, showbiz, hollywood
- (33) **sports_competition_games**: competition, tournament, match, games, game, contest, training, initiation, challenge, survive, survival, fight, fighting, battle, arena, victory, win, winner, lose, losing, tribute, reality, tv, faction, selection, crown, prince, princess, sports, team, rugby, coach, practice, race, olympics, run, running
- (34) **parenting_babycare**: baby, babies, newborn, infant, toddler, child, children, parent, parents, mom, moms, mother, mothers, dad, dads, father, fathers, crib, bassinet, nursery, stroller, carseat, diaper, diapers, cloth, diapering, wipes, bottle, bottles, formula, breastfeeding, breastmilk, latch, pumping, milk, feeding, feedings, sleep, sleeping, nap, naps, sleep-training, swaddle, pacifier, teething, colic, burp, burping, bath, bath-time, pajamas, pajama, potty, potty-training, weaning, solids, puree, homemade, food, vitamin, supplements, doctor, pediatrician, vaccines, immunization, growth, milestones, month-by-month, screen-time, tablet, tv, apps, safety, childproof, attachment, parenting
- (35) **work_labor_service_industry**: work, job, jobs, worker, workers, labor, shift, wages, pay, paycheck, salary, tips, tipping, minimum, wage, poverty, poor, working, class, waitress, waiter, server, maid, housekeeper, housekeeping, cleaning, janitor, nursing, home, aide, retail, wal-mart, walmart, big, box, hotel, motel, bellhop, bellman, valet, parking, front, desk, concierge, guest, guests, customer, customers, service, industry, manager, boss, coworker, abuse, overworked, overtime, schedule, rent, lodging, cheap, rooms, survival, stratagem, union, strike

- (36) **political_history_presidency**: president, presidency, white, house, oval, office, campaign, election, reagan, ronald, governor, california, hollywood, cold, war, soviet, iron, curtain, assassin, assassination, bullet, hinckley, secret, service, press, conference, speech, statecraft, policy, congress, senate, representatives, democrat, republican, conservative, liberal, communism, capitalism, diplomacy, summit, gorbachev, union, history, biography
- (37) **cult_high_control_religion**: cult, cults, sect, sects, high-control, indoctrination, brainwashing, coercion, control, obedience, punishment, discipline, shunning, isolation, escape, defect, defection, insider, whistleblower, secrecy, ritual, rituals, leader, leadership, celebrity, recruitment, auditing, e-meter, scientology, sea, org, david, miscavige, church, organization, faith, religion, belief, doctrine, dogma, apostate, freedom, trauma, family
- (38) **comics_superheroes_graphic_novels**: comic, comics, graphic, novel, manga, volume, issue, issues, panel, panels, superhero, superheroes, hero, villain, vigilante, mask, cape, gotham, batman, joker, dark, knight, returns, dc, marvel, origin, story, reboot, crossover, arc, series, collecting, collector, edition, illustrated, art, ink, pencil, storyboard
- (39) **art_history_visual_arts**: art, history, artist, artists, painting, paintings, sculpture, sculptures, drawing, drawings, sketch, sketches, portrait, portraits, landscape, still, life, museum, gallery, exhibition, renaissance, baroque, impressionism, modern, modernism, abstract, cubism, surrealism, icon, icons, fresco, canvas, oil, watercolor, acrylic, architecture, cathedral, design, aesthetics, criticism, critique, artwork, masterpiece, masters
- (40) **christian_rapture_apocalypse**: rapture, tribulation, antichrist, apocalypse, apocalyptic, revelation, revelations, prophecy, prophecies, endtimes, end-times, judgment, judgement, salvation, saved, unsaved, believers, unbelievers, church, pastor, preacher, bible, scripture, heaven, hell, angels, demon, demons, disappear, disappearance, vanish, vanished, left, behind, plague, mark, beast, armageddon
- (41) **satire_parody_comedy**: satire, satirical, parody, spoof, lampoon, comedy, comic, humor, humour, hilarious, witty, wit, absurd, ridiculous, farce, irony, ironic, mock, mocking, sendup, tongue-in-cheek, pastiche, sarcasm, sarcastic, bizarre, weird, wacky, quirky, dark, laugh, laughing, jokes
- (42) **theater_stage_play**: play, stage, theatre, theater, drama, dramatic, scene, scenes, act, acts, dialogue, monologue, character, characters, audience, performance, script, screenplay, director, actors, actresses, broadway, pulitzer, critics, circle, award, production, rehearsal, opening, night
- (43) **immortality_eternal_life**: immortal, immortality, eternal, life, forever, never, die, deathless, ageless, age, youth, spring, magic, water, drink, drinking, curse, blessing, blessed, doomed, secret, live, living, regret, consequence, mortality, mortal, human, choice, time
- (44) **spies_espionage_thriller**: spy, spies, espionage, agent, agents, secret, mission, assignment, operation, operative, smersh, 007, bond, james, harlem, caribbean, everglades, jazz, joints, criminal, heavy, hitter, threat, assassin, assassination, weapon, weapons, gun, guns, chase, pursuit, cover, undercover, surveillance, interrogation
- (45) **haunted_objects_vehicles**: haunted, cursed, possessed, object, objects, artifact, artifacts, doll, mirror, house, car, cars, automobile, plymouth, fury, engine, garage, drive, driving, headlights, radio, rock, roll, detroit, accident, crash, blood, revenge, jealousy, obsession, evil
- (46) **business_entrepreneurship**: business, company, companies, startup, start-up, entrepreneur, entrepreneurship, founder, founding, ceo, executive, manager, management, leadership, strategy, strategic, product, products, market, marketing, sales, branding, brand, customer, customers, growth, scale, scaling, revenue, profit, loss, losses, budget, budgets, funding, investor, investors, venture, capital, vc, pitch, deal, negotiation, contract, contracts, team, teams, culture, innovation, disruption, competition, competitive, industry, industries, supply, demand, operations, logistics, metrics, kpi, analytics
- (47) **finance_investing_money**: money, wealth, rich, poor, income, salary, wages, budget, saving, savings, debt, loan, loans, credit, mortgage, interest, compound, investment, investments, investing, investor, investors, portfolio, asset, assets, stocks, stock, bond, bonds, equity, index, fund, funds, eff, dividend, dividends, market, markets, bull, bear, risk, return, returns, valuation, value, cash, cashflow, inflation, recession, economy, economics, finance, financial, retirement, pension, tax, taxes, insurance
- (48) **health_fitness_nutrition**: health, healthy, fitness, fit, workout, workouts, exercise, training, gym, running, run, yoga, pilates, strength, cardio, endurance, muscle, weight, weights, fat, calories, calorie, diet, nutrition, nutritional, protein, carbs, carbohydrate, fiber, vitamin, vitamins, supplement, supplements, sleep, stress, wellness, mind, body, metabolism, hydration, water, meal, meals, recipe, recipes, cook, cooking, kitchen
- (49) **psychology_self_help_mindfulness**: psychology, psychological, mind, brain, behavior, behaviour, habit, habits, motivation, discipline, willpower, mindset, self, help, self-help, confidence, anxiety, depression, stress, trauma, therapy, therapist, counseling, counselling, healing, growth, personal, development, mindfulness, meditation, breath, breathing, gratitude, compassion, empathy, relationships, communication, boundaries, resilience, purpose, meaning, happiness, joy, focus, productivity, procrastination
- (50) **cooking_food_recipes**: cook, cooking, kitchen, recipe, recipes, ingredients, ingredient, bake, baking, roast, grill, saute, sauté, boil, simmer, fry, fried, oven, stove, pan, pot, knife, spice, spices, herb, herbs, salt, pepper, sugar, flour, butter, oil, garlic, onion, tomato, cheese, bread, cake, dessert, chocolate, soup, stew, salad, sauce, pasta, rice, beans, meat, chicken, beef, pork, fish, seafood, vegetarian, vegan, gluten, nutrition, meal, meals, menu
- (51) **technology_programming_computers**: technology, tech, computer, computers, software, hardware, programming, programmer, code, coding, developer, development, debug, bug, algorithm, algorithms, data, database, databases, network, networks, internet, web, website, app, apps, mobile,

server, cloud, security, encryption, cryptography, ai, ml, machine, learning, neural, model, models, training, dataset, api, open-source, opensource, linux, python, java, javascript, terminal, command, git, github

- (52) **popular_science_nature**: science, scientific, physics, chemistry, biology, genetics, dna, evolution, neuroscience, astronomy, cosmos, universe, planet, planets, space, ecology, environment, climate, energy, atoms, molecule, molecules, experiment, experiments, research, scientist, scientists, theory, theories, evidence, data, discovery, discoveries, laboratory, lab, nature, wildlife, species

Bellow we enumerate the Topic-dictionary based on MovieLens:

- (1) **crime_underworld**: gangster, mob, mafia, cartel, hitman, assassin, crime boss, racketeering, drug deal, smuggling, black market, undercover, informant, witness, shootout, betrayal, corrupt, extortion, loan shark, criminal empire, street gang, organized crime, revenge, vigilante, fugitive, manhunt, prison, parole, robbery, kidnapping, ransom
- (2) **heists_cons**: heist, bank robbery, vault, safecracker, con artist, scam, grift, swindle, identity theft, forgery, counterfeit, inside job, getaway, crew, stakeout, double-cross, score, loot, ransack, hostage, ransom, hijack, blackmail, stolen art, diamond, casino, pickpocket, hustler, underworld contact
- (3) **mystery_investigation**: detective, investigation, case, clue, suspect, interrogation, forensics, evidence, crime scene, murder mystery, whodunit, missing person, disappearance, cold case, alibi, witness statement, private investigator, police chief, serial killer, profiling, autopsy, trail, cover-up, conspiracy, secret, twist ending
- (4) **spy_espionage**: spy, espionage, secret agent, double agent, CIA, MI6, KGB, intelligence, surveillance, wiretap, classified, black ops, covert, infiltrate, counterintelligence, asset, handler, mole, defection, safehouse, passport, interrogation, torture, diplomatic, nuclear, bioweapon, cyberterror, mission, extraction
- (5) **war_military**: war, battle, soldier, army, marines, navy, air force, mission, frontline, trench, sniper, bomber, invasion, resistance, occupation, prisoner of war, general, colonel, unit, comrades, boot camp, veteran, homecoming, propaganda, spy plane, wartime romance, medic, sacrifice, ceasefire, atrocity
- (6) **survival_disaster**: survival, stranded, shipwreck, plane crash, wilderness, storm, hurricane, earthquake, tsunami, volcano, fire, flood, rescue, evacuation, catastrophe, doomsday, asteroid, pandemic, outbreak, quarantine, scarcity, rationing, shelter, search party, last survivors, endurance, hypothermia, dehydration, sos
- (7) **martial_arts_combat**: martial arts, kung fu, karate, dojo, master, disciple, tournament, sparring, fight, brawl, duel, sword, blade, samurai, ninja, warlord, training montage, honor, revenge, clan, warrior, champion, gladiator, combat, assassin, standoff, boss fight, arena, blood feud
- (8) **quest_adventure**: quest, journey, expedition, treasure, map, artifact, ruins, temple, ancient, curse, legend, myth, escape, chase, hideout, lost city, jungle, desert, island, mountain, cave, ship, voyage, explorer, pirate, captain, sailing, tribe, mystery box, peril
- (9) **space_aliens**: space, spaceship, astronaut, planet, galaxy, starship, orbit, mission control, alien, extraterrestrial, invasion, first contact, abduction, UFO, colony, terraform, wormhole, hyperspace, space station, laser, ray, space marine, cosmic, signal, telemetry, asteroid belt, moon base, Martian, exoplanet, mothership
- (10) **time_travel_paradox**: time travel, time loop, paradox, timeline, alternate history, future, past, temporal agent, chrononaut, rewind, foresee, predestination, butterfly effect, grandfather paradox, memory reset, déjà vu, prophecy, fate, destiny, reality shift, multiverse, dimension, portal, anomaly, time machine, retrocausality, looping day, foretold, identity twist
- (11) **robots_AI**: robot, android, cyborg, AI, artificial intelligence, machine, sentient, algorithm, hack, hacker, cyber, virtual reality, simulation, upload, consciousness, dystopian tech, surveillance state, drone, automaton, malfunction, singularity, clone, genetic, bioengineering, lab, experiment, nanotech, firmware, mainframe, digital
- (12) **dystopia_apocalypse**: dystopia, post-apocalyptic, collapse, ruins, wasteland, mutants, scavenger, bunker, radiation, nuclear, totalitarian, regime, rebellion, uprising, oppression, underground, blackout, curfew, ration, resistance, propaganda, chosen one, civil war, martial law, contagion, end times, fallout, last city, survivors
- (13) **magic_myth**: magic, wizard, witch, spell, curse, prophecy, chosen one, enchanted, sorcerer, warlock, alchemy, potion, rune, artifact, mythology, gods, demigod, oracle, legendary, dragon, giant, troll, fairy, elf, dwarf, quest, kingdom, throne, knight, castle
- (14) **vampires_werewolves**: vampire, blood, immortal, undead, coven, fangs, werewolf, lycan, pack, moon, transformation, hunter, stake, curse, darkness, night, forbidden love, ancient evil, seduction, supernatural, witch, cursed lineage, bite, ritual, haunted, crypt, catacombs, curse breaking, bloodline, immortality
- (15) **ghosts_hauntings**: ghost, haunted, spirit, poltergeist, possession, exorcism, paranormal, medium, seance, cursed house, apparition, afterlife, demon, ritual, occult, nightmare, creepy, whisper, shadow, screams, hotel room, basement, graveyard, mystic, curse, hallucination, superstition, urban legend, haunting past, restless
- (16) **monsters_creature_horror**: monster, creature, beast, mutant, parasite, infection, experiment gone wrong, lab creature, predator, hunt, bloodthirsty, killer, cannibal, slasher, serial killer, zombie, undead horde, outbreak, body horror, were-beast, giant insect, sea monster, alien predator, survival horror, stalker, terror, panic, hide, trapped, final girl
- (17) **romance_heartbreak**: love, romance, affair, cheating, breakup, heartbreak, wedding, engagement, proposal, forbidden love, love triangle, jealousy, reunion, second chance, long distance, letters, destiny, soulmate, passion, seduction, dating, flirtation, honeymoon, divorce, widow, mourning, first love, unrequited, commitment, intimacy

- (18) **family_parenting**: family, parents, mother, father, siblings, daughter, son, guardian, custody, divorce, stepfamily, orphan, adoption, home, coming home, inheritance, legacy, generations, family secret, caregiving, illness, funeral, reconciliation, dysfunctional, holiday gathering, raising a child, teenage, responsibility, family business, protect
- (19) **friendship_coming-of-age**: friendship, best friends, growing up, coming-of-age, teen, high school, summer, first kiss, bullies, outsider, misfit, self-discovery, identity, graduation, college, young adult, road trip, band, party, prom, sports team, new kid, mentor, peer pressure, rebellion, dreams, small town, nostalgia, childhood, rite of passage, found family
- (20) **comedy_satire**: comedy, satire, spoof, parody, awkward, misunderstanding, farce, fish out of water, odd couple, schemes, prank, ridiculous, absurd, slapstick, rom-com, wedding chaos, road comedy, buddy comedy, workplace comedy, stand-up, celebrity, showbiz, small-time, hustle, quirky, deadpan, mockumentary, midlife crisis, family comedy, culture clash
- (21) **courtroom_justice**: court, trial, lawyer, attorney, judge, jury, testimony, verdict, appeal, lawsuit, evidence, cross-examination, legal thriller, wrongful conviction, innocent, prison, death penalty, corruption, cover-up, whistleblower, civil rights, scandal, investigation, prosecution, defense, plea deal, witness protection, case files, moral dilemma, justice
- (22) **politics_power**: politics, mayor, president, campaign, election, senate, parliament, dictator, coup, revolution, state, government, bureaucracy, corruption, spin, press, journalist, whistleblower, conspiracy, diplomacy, treaty, war room, policy, protest, activist, ideology, power struggle, assassination, propaganda, regime
- (23) **race_class_inequality**: racism, segregation, prejudice, class, poverty, wealth, privilege, immigrant, refugee, oppression, labor, strike, hunger, urban, ghetto, community, justice, civil rights, protest, discrimination, identity, belonging, marginalized, survival, education, jobless, struggle, neighborhood, family duty, hope
- (24) **work_ambition**: career, ambition, success, failure, promotion, boss, office, workplace, intern, deadline, competition, fame, celebrity, artist, writer, musician, actor, producer, startup, deal, betrayal, rivalry, dream job, training, performance, pressure, burnout, mentor, second act, reinvention
- (25) **music_performance**: music, band, singer, concert, tour, rock, jazz, classical, composer, piano, guitar, drummer, record, album, stage, spotlight, dance, ballet, opera, show, theater, audition, festival, talent, fame, backstage, groupies, breakup tour, encore, soundtrack
- (26) **sports_competition**: boxing, wrestling, race, track, olympics, football, basketball, baseball, coach, training, championship, tournament, underdog, comeback, injury, rivalry, team, locker room, draft, agent, betting, gambling, marathon, fighter, knock-out, ring, stadium, victory, defeat, discipline
- (27) **historical_epics**: historical, period, empire, royalty, king, queen, palace, court, dynasty, reign, colonial, plantation, aristocrat, noble, serf, revolution, war of independence, victorian, medieval, renaissance, ancient, gladiator, conquest, uprising, battles, palace intrigue, succession, landed estate, family lineage, chronicle
- (28) **western_frontier**: western, cowboy, outlaw, sheriff, frontier, ranch, saloon, gunfight, duel, posse, bounty, trail, wagon, settlers, gold rush, train robbery, desert town, horse, marshal, bandit, revenge, lawless, border, territory, indian, cattle drive, showdown, wanted poster, homestead, range
- (29) **noir_psychological**: noir, psychological, paranoia, obsession, delusion, unreliable narrator, identity, dream, hallucination, mind game, therapy, trauma, memory, guilt, confession, double life, dark secret, moral ambiguity, femme fatale, betrayal, loneliness, alienation, despair, existential, compulsion, meltdown, introspection, fatalism, spiral, black-and-white
- (30) **animation_kids**: animation, cartoon, toy, talking animal, princess, fairytale, storybook, magical friend, adventure for kids, school, playground, family-friendly, sing-along, sidekick, villain, lesson, friendship, imagination, holiday, cute, animal companions, coming home, child hero, dreamland, storybook world, rescue mission, playtime, kidnapped pet, happy ending, wholesome
- (31) **animals_nature**: animals, dog, cat, horse, wildlife, jungle, forest, ocean, whale, shark, bear, lion, zoo, ranger, habitat, conservation, environment, nature documentary, ecosystem, pet, trainer, rescue animal, migration, mountain, island, stormy sea, expedition, safari, pack, instinct

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